Game Theory in Electricity Markets

Week 4 – Price of Anarchy

Pure Nash equilibrium

Definition. An action profile $a^* = (a_1^*, \ldots, a_n^*) \in A$ is a *Nash equilibrium* if no player can lower their own cost by unilaterally deviating:

$$c_i(a_i^*, a_{-i}^*) \leq c_i(a_i, a_{-i}^*)$$

for every player $i \in N$ and every $a_i \in A_i$.

ε-Nash equilibrium

Definition. An action profile $a^* = (a_1^*, \ldots, a_n^*) \in A$ is an *ε-Nash equilibrium* ($\varepsilon \geq 0$) if no player can lower their own cost by more than ε by unilaterally deviating:

$$c_i(a_i^*, a_{-i}^*) \leq c_i(a_i, a_{-i}^*) + \varepsilon$$

for every player $i \in N$ and every $a_i \in A_i$.

For ε = 0 this is a pure Nash equilibrium.

Mixed Nash equilibrium

Definition. A mixed profile $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a *mixed Nash equilibrium* if no player can lower their expected cost by unilaterally deviating:

$$c_i(\sigma_i^*, \sigma_{-i}^*) \leq c_i(\sigma_i, \sigma_{-i}^*)$$

for every player $i \in N$ and every $\sigma_i \in \Delta(A_i)$.

A pure Nash equilibrium is the special case where every σ_i^* puts probability 1 on a single action.

ε-mixed Nash equilibrium

Definition. A mixed profile $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is an *ε-mixed Nash equilibrium* ($\varepsilon \geq 0$) if no player can lower their expected cost by more than ε by unilaterally deviating:

$$c_i(\sigma_i^*, \sigma_{-i}^*) \leq c_i(\sigma_i, \sigma_{-i}^*) + \varepsilon$$

for every player $i \in N$ and every $\sigma_i \in \Delta(A_i)$.

For ε = 0 this is a mixed Nash equilibrium.

Correlated equilibrium

Definition. A distribution $\pi \in \Delta(A)$ over the joint action set A is a *correlated equilibrium* if no player can lower their expected cost by deviating from a privately recommended action:

$$E_{a\sim\pi}\,[c_i(a) \mid a_i] \;\leq\; E_{a\sim\pi}\,[c_i(a_i', a_{-i}) \mid a_i]$$

for every player $i \in N$ and every $a_i, a_i' \in A_i$.

A correlated equilibrium π that can be written as an independent (product) distribution, $\pi(a) = \sigma_1(a_1) \cdots \sigma_n(a_n)$, is a mixed Nash equilibrium.

ε-correlated equilibrium

Definition. A distribution $\pi \in \Delta(A)$ over the joint action set A is an *ε-correlated equilibrium* ($\varepsilon \geq 0$) if no player can lower their expected cost by more than ε by deviating from a privately recommended action:

$$E_{a\sim\pi}\,[c_i(a) \mid a_i] \;\leq\; E_{a\sim\pi}\,[c_i(a_i', a_{-i}) \mid a_i] + \varepsilon$$

for every player $i \in N$ and every $a_i, a_i' \in A_i$.

For ε = 0 this is a correlated equilibrium.

Coarse correlated equilibrium

Definition. A distribution $\pi \in \Delta(A)$ is a *coarse correlated equilibrium* if no player can lower their expected cost by an unconditional deviation to a fixed action:

$$E_{a\sim\pi}[c_i(a)] \leq E_{a\sim\pi}[c_i(a_i', a_{-i})]$$

for every player $i \in N$ and every $a_i' \in A_i$.

Every correlated equilibrium is a coarse correlated equilibrium.

ε-coarse correlated equilibrium

Definition. A distribution $\pi \in \Delta(A)$ is an *ε-coarse correlated equilibrium* ($\varepsilon \geq 0$) if no player can lower their expected cost by more than ε by an unconditional deviation to a fixed action:

$$E_{a\sim\pi}[c_i(a)] \leq E_{a\sim\pi}[c_i(a_i', a_{-i})] + \varepsilon$$

for every player $i \in N$ and every $a_i' \in A_i$.

For ε = 0 this is a coarse correlated equilibrium. Every ε-correlated equilibrium is an ε-coarse correlated equilibrium.

How to measure inefficiency of a market

Equilibrium concept of our choice

$$Price\ of\ Anarchy = max \frac{Cost\ of\ the\ emergent\ game\ outcome}{Minimum\ cost}$$

$Price\ of\ Anarchy$ = 1, society pays **as little as possible** in any outcome

$Price\ of\ Anarchy$ = 2, we pay **twice as much** in a realistic outcome

Price of Anarchy

Level 3 — Maximize the cost ratio over all such settings

Level 2 — Players maximize revenue

Level 1 — The operator clears the market, given fixed bids

The market clearing in itself is an optimization problem!

Price of Anarchy

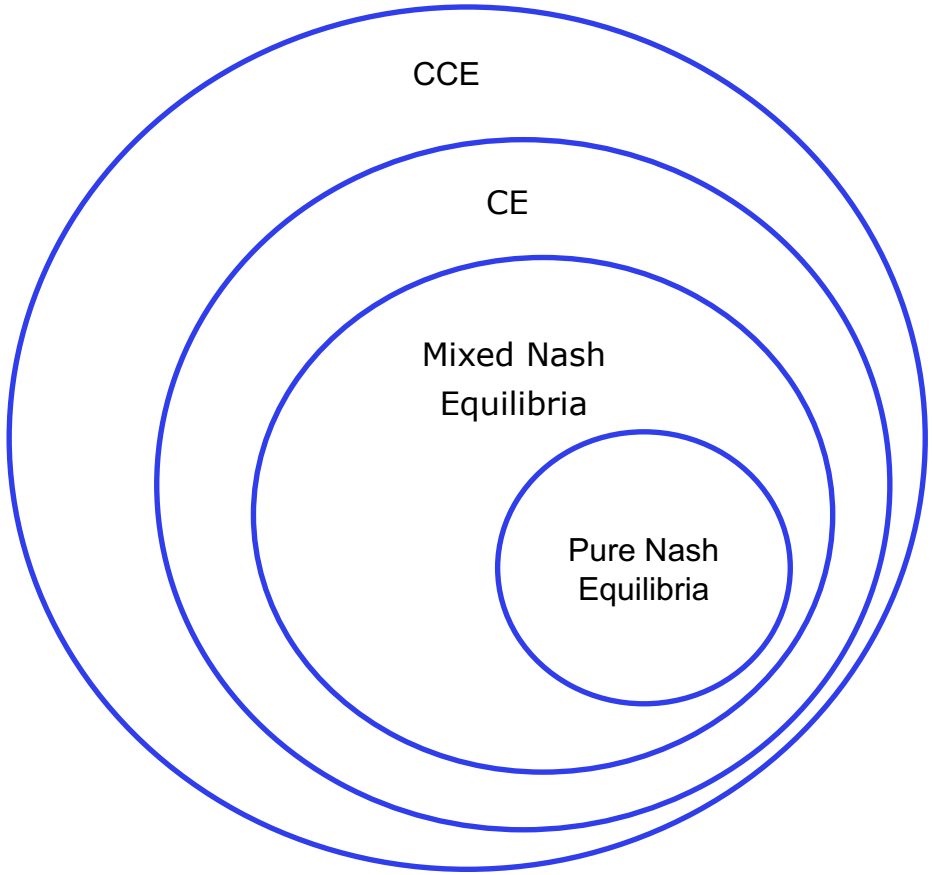
Level 3 — Maximize the cost ratio over all such settings

Set of Equilibria

Price of Anarchy

$$\text{PoA}_{\text{PNE}} \leq \text{PoA}_{\text{MNE}} \leq \text{PoA}_{\text{CE}} \leq \text{PoA}_{\text{CCE}}$$

For a fixed ε: $$\text{PoA}_{\varepsilon\text{-PNE}} \leq \text{PoA}_{\varepsilon\text{-MNE}} \leq \text{PoA}_{\varepsilon\text{-CE}} \leq \text{PoA}_{\varepsilon\text{-CCE}}$$



Equilibria sets of finite games

Γ_{PNE}: possibly empty, but subset of finite set

Γ_{MNE}: not empty under weak conditions, finitely many choices for the support of each players strategy

Γ_{CE}: defined by $O(|A_i|^2 \cdot n)$ linear constraints

Γ_{CCE}: defined by $O(|A_i| \cdot n)$ linear constraints

→ For finite action spaces we can determine all these sets, although some in exponential time

Infinite case

Γ_{PNE}: possibly empty, could lay anywhere

Γ_{MNE}: not empty under weak conditions, but no structure

Γ_{CE}: defined by infinitely many linear constraints

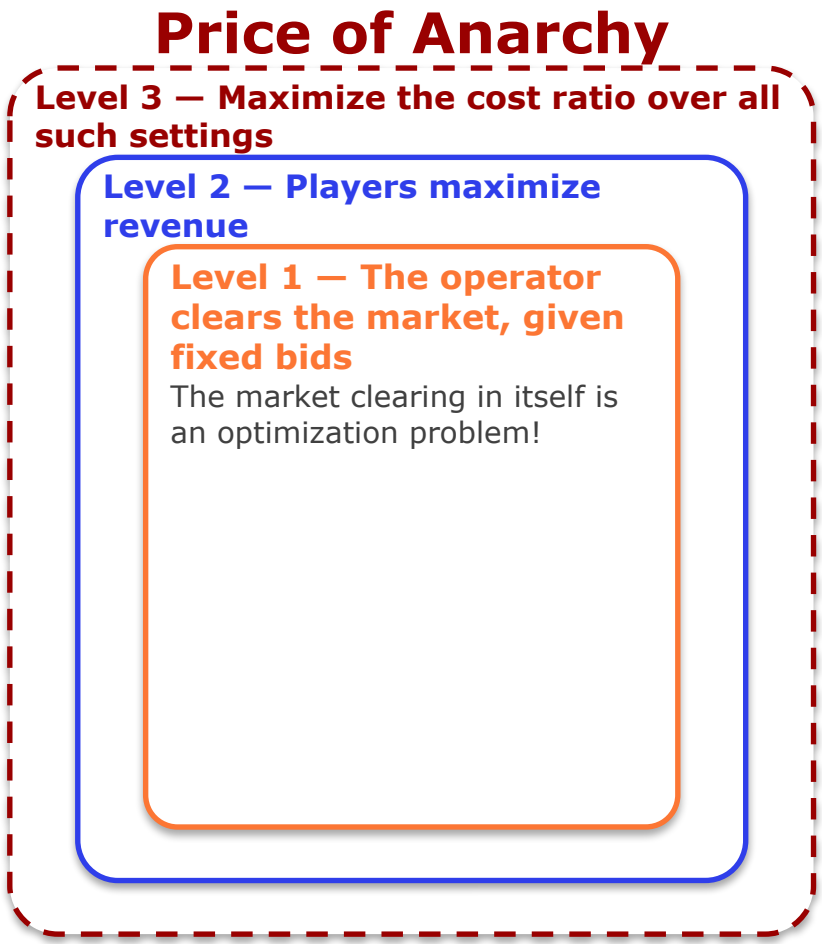
Γ_{CCE}: defined by infinitely many linear constraints

→ For infinite action spaces we generally can't derive any of these sets

Any ideas how to derive PoA for infinite action spaces?

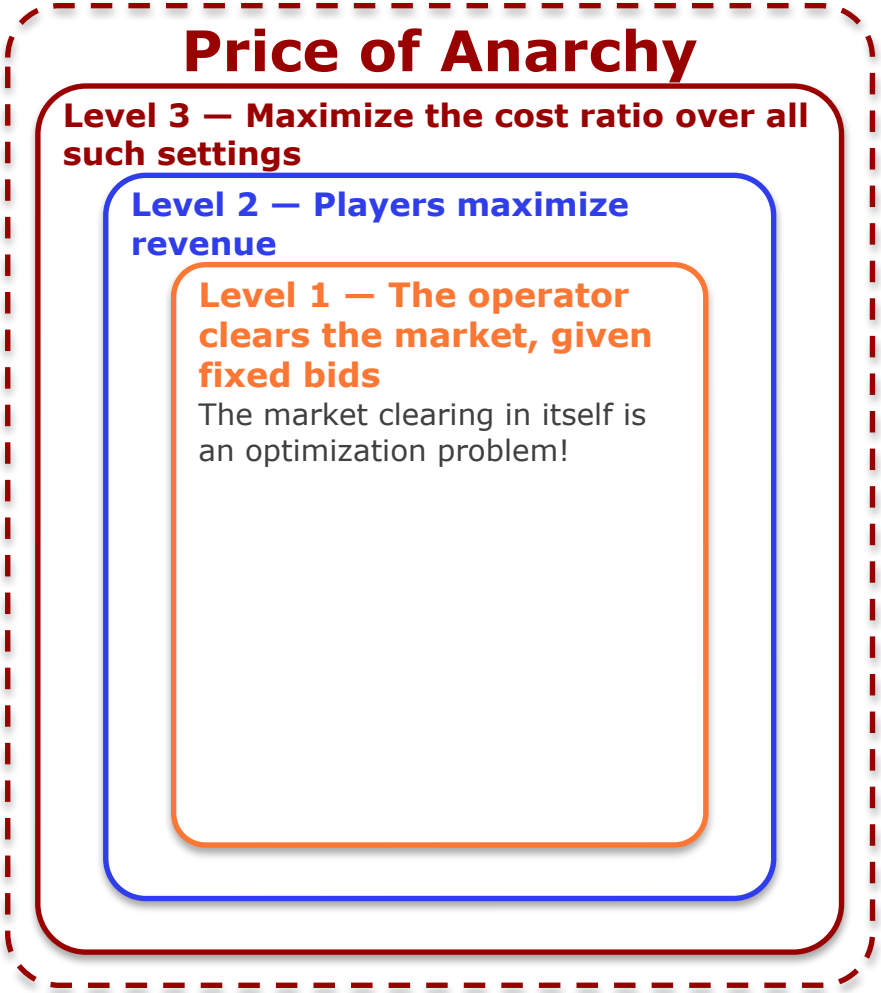
Lower bounds

- Instead of solving the outer problem just consider its feasibility problem
- In other words, find *some* equilibria
- The maximum inefficiency over those is a lower bound on PoA
- Problem: we don’t know how far we are off



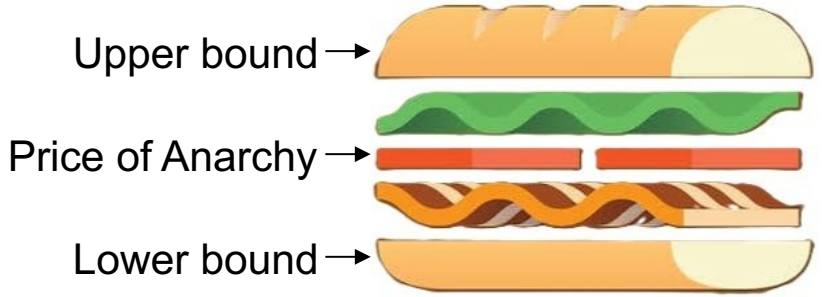
Upper bounds

- Instead of solving the outer problem find an upper bound on its solution
- Find a set that contains all equilibria and for which you can make a statement about the maximal social cost
- Simple example: maximizing social cost over the whole joint action space induces a PoA bound
- Problems: We do not know how far we are off. Also finding an upper bound might be easy, but finding a good one is a whole new problem



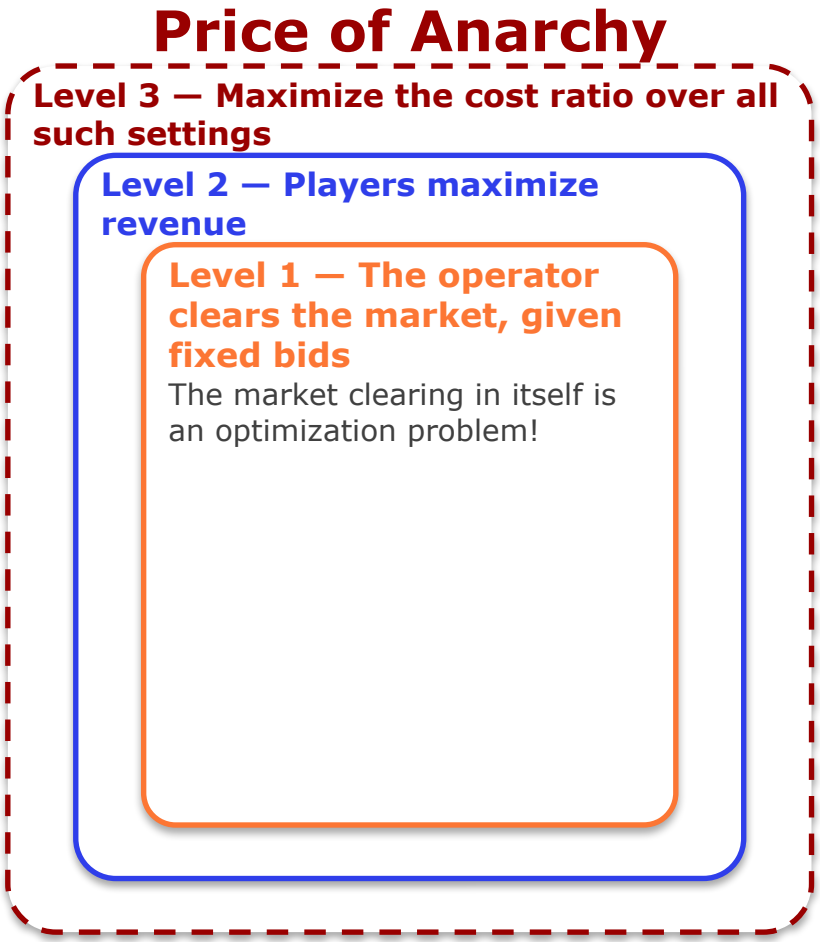
PoA sandwich

- Combining an upper and a lower bound boxes the PoA value in
- If the bounds are close enough, we essentially have a numerical value for PoA
- We don’t even need the lower bound if we can make a statement about the quality of the upper bound
- Problems: we still have no way to calculate the bounds, especially the upper one



Finding lower bounds

- Use heuristics to find equilibria
- If the heuristic has some game theoretic meaning (for example search for NE via BR) we have interpretation of the result that are more then “lower bound on PoA”
- If interested in quality of the upper bound instead of a valid lower bound one might also be okay with ”just” finding ε-equilibria
- These one can often find by discretizing the game and calculating the PoA over them



Questions?

Upper bounds via (λ,μ)-smoothness

Definition. A cost-minimization game with social cost C is (λ, μ)-*generalized smooth* (λ > 0, μ < 1) if for all a, a′ ∈ A:

$$\left[\Sigma_i\, c_i(a'_i,\, a_{-i}) \;-\; \Sigma_i\, c_i(a)\right] \;+\; C(a) \;\leq\; \lambda\, C(a') \;+\; \mu\, C(a)$$

Theorem (Chandan, Paccagnan, Marden). If a game is (λ, μ)-generalized smooth, then

$$PoA \;\leq\; \lambda \,/\, (1 - \mu)$$

Proof of the PoA bound

Proof. We prove it for pure Nash equilibria, but the bound stays true for all equilibrium concepts we have seen so far (exercise).

Let a be a pure Nash equilibrium and a* an optimal profile:

$$C(a) \leq C(a) + \Sigma_i [\, c_i(a^*_i, a_{-i}) - c_i(a) \,]$$

a is a Nash equilibrium: every bracket ≥ 0

$$\leq \lambda\, C(a^*) + \mu\, C(a)$$

generalized smoothness with a′ = a*

$$C(a) / C(a^*) \leq \lambda / (1 - \mu)$$

rearranging (1 − μ) C(a) ≤ λ C(a*), using μ < 1

Upper bounds via smoothness

- The idea and the proof are simple (short and does not use any advanced concepts)
- Still, it is one of the most used approaches to bound/calculate PoA
- If the game structure is simple enough, one may find $\boldsymbol{\lambda}, \boldsymbol{\mu}$ via theoretical considerations
- Otherwise, the best smoothness bound is given via finding the solution of the following program

$$\inf_{\lambda>0,\ \mu<1} \quad \lambda / (1 - \mu)$$

$$\text{s.t.} \quad \Sigma_i\, c_i(a'_i, a_{-i}) - \Sigma_i\, c_i(a) + C(a) \leq \lambda\, C(a') + \mu\, C(a) \qquad \forall\ a, a' \in A$$

Change of variables

With $\rho = \lambda / (1 - \mu)$ and $v = 1 / (1 - \mu)$, i.e. $\lambda = \rho / v$ and $\mu = 1 - 1 / v$, the program becomes linear in (ρ, v).

$$\inf_{\lambda>0,\ \mu<1} \quad \lambda / (1 - \mu)$$

$$\text{s.t.} \quad \Sigma_i\, c_i(a'_i, a_{-i}) - \Sigma_i\, c_i(a) + C(a) \leq \lambda\, C(a') + \mu\, C(a) \quad \forall\ a, a' \in A$$

change of variables →

$$\inf_{\rho>0,\ v>0} \quad \rho$$

$$\text{s.t.} \quad C(a) + v\,[\,\Sigma_i\, c_i(a'_i, a_{-i}) - \Sigma_i\, c_i(a)] \leq \rho\, C(a') \quad \forall\ a, a' \in A$$

Fixed socially optimal profile

Remark. The proof only uses a' = a*. Therefore, the bound also holds if we only compare against a fixed socially optimal profile a*, which gives the programs:

$$\inf_{\lambda>0,\ \mu<1} \quad \lambda / (1 - \mu)$$

$$\text{s.t.} \quad \Sigma_i\, c_i(a^*_i, a_{-i}) - \Sigma_i\, c_i(a) + C(a) \leq \lambda\, C(a^*) + \mu\, C(a) \quad \forall\, a \in A$$

change of variables →

$$\inf_{\rho>0,\ v>0} \quad \rho$$

$$\text{s.t.} \quad C(a) + v\,[\,\Sigma_i\, c_i(a^*_i, a_{-i}) - \Sigma_i\, c_i(a)] \leq \rho\, C(a^*) \quad \forall\, a \in A$$

Smoothness short recap

- Smoothness gives us a certified upper bound via a linear program with $|A_i|^n$ constraints, that means it has infinitely many constraints if A_i is infinite
- It does not tell us how to solve this program nor how good this bound is
- The idea is to essentially bound the social cost over the whole joint action space but to add a correction term such that this maximization cannot be driven by a very costly but completely instable joint action

$$\left[\sum_i c_i(a'_i, a_{-i}) - \sum_i c_i(a)\right] + C(a) \leq \lambda C(a') + \mu C(a)$$

Fixed socially optimal profile revisited

Theorem (Strauch, Kamgarpour, Tsaousoglou, still not yet out). Fix A* as the set of socially optimal joint action profiles. Then the following programs yield an upper bound to the Price of Anarchy that is weakly tighter than the one of generalized smoothness.

$$\inf_{\lambda>0,\ \mu<1,\ a^* \in A^*} \quad \lambda / (1 - \mu)$$

$$\text{s.t.} \quad \Sigma_i\, c_i(a^*_i, a_{-i}) - \Sigma_i\, c_i(a) + C(a) \leq \lambda\, C(a^*) + \mu\, C(a) \quad \forall\, a \in A$$

change of variables →

$$\inf_{\rho>0,\ v>0,\ a^* \in A^*} \quad \rho$$

$$\text{s.t.} \quad C(a) + v\,[\,\Sigma_i\, c_i(a^*_i, a_{-i}) - \Sigma_i\, c_i(a)] \leq \rho\, C(a^*) \quad \forall\, a \in A$$

The problem (what I do the other 4.5 days a week): the set A* does generally not have any nice properties.

The solution: in electricity markets a relevant subset of A* can be parameterized (next week)


DTU